Brian Urbina, Shantonu Mandal, Alex McGee
WES 237C

Project 3 Report: Discrete Fourier Transform

Question 1 - DFT32 Baseline

Part (a)

Yes, changing the accuracy (bit width) of a custom CORDIC core would impact the resource usage. Increasing the number of bits would use more resources. A DFT32 needs 32 x 32 = 1024 sin/cos calculations. So any additional CORDIC latency will be multiplied across a single transform - thus lowering the DFT throughput. Also, any error in the CORDIC will carry into the DFT output.

Brian Urbina, Shantonu Mandal, Alex McGee
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Question 2 - DFT32 Table Lookup

Part (a)

	Baseline (Q1)	**Table Lookup**	**Change**
Latency (clock cycles)	54884 - 55908	257	⬇️ ~214x
Initiation Interval (II) (clock cycles)	54885 - 55909	258	⬇️ ~213x
Throughput (DFT/s)	1790 - 1820	387,000	⬆️ ~214x
BRAM	2	66	⬆️ 33x
DSP	53	640	⬆️ ~12x
FF	5275	59208	⬆️ ~11x
LUT	9111	91730	⬆️ ~10x

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Question 3 - DFT32 Interface Change

Part (a)

Accepting the output array as an input to dft() (it will translate to a separate input port/stream in DFT core) avoids doing an in-place DFT. So, the HLS function no longer reads and writes to the same location. And helps do away with the temporary buffer and copying back the output loop.

With separate arrays, we can partition each one independently - getting a lower II. It also reduces latency since we can get rid of the output copy loop at the end of the baseline version.

Part (b)

	Table Lookup (in-place) (Q2)	**Interface Change**	**Change %**
Latency (clock cycles)	257	221	⬇️ 14%
Initiation Interval (II) (clock cycles)	258	222	⬇️ ~14%
Throughput (DFT/s)	387,000	450,500	⬆️ 16%
BRAM	66	64	⬇️ 3%
DSP	640	640	0
FF	59208	59189	⬇️ ~0.03%
LUT	91730	91521	⬇️ ~0.2%

Part (c)

With the interface change, latency and II dropped by 14% (and throughput increased, correspondingly). This is likely due to the output copy loop removal.
Resource usage stayed roughly the same.

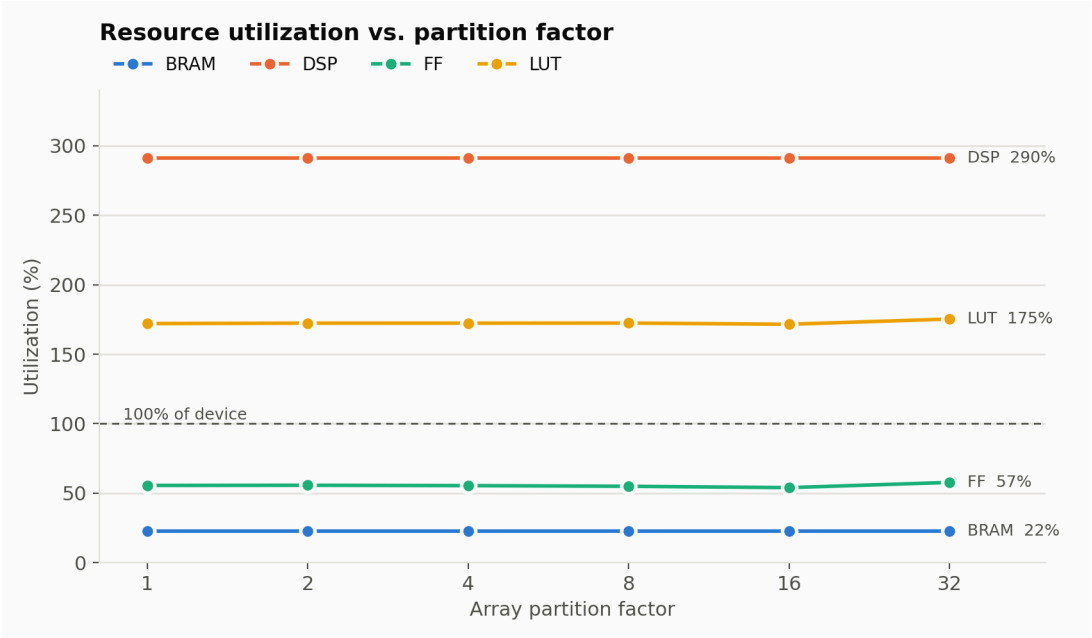
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Question 4 - DFT32 Array Partitioning

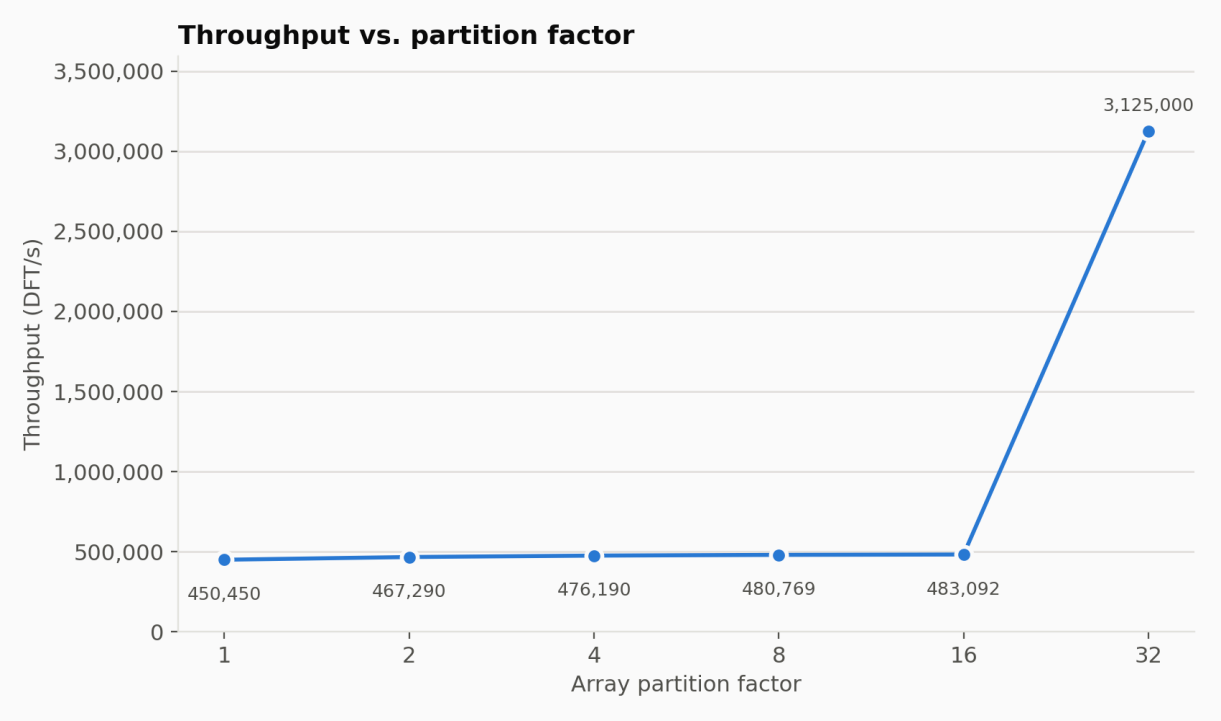
Part (a)

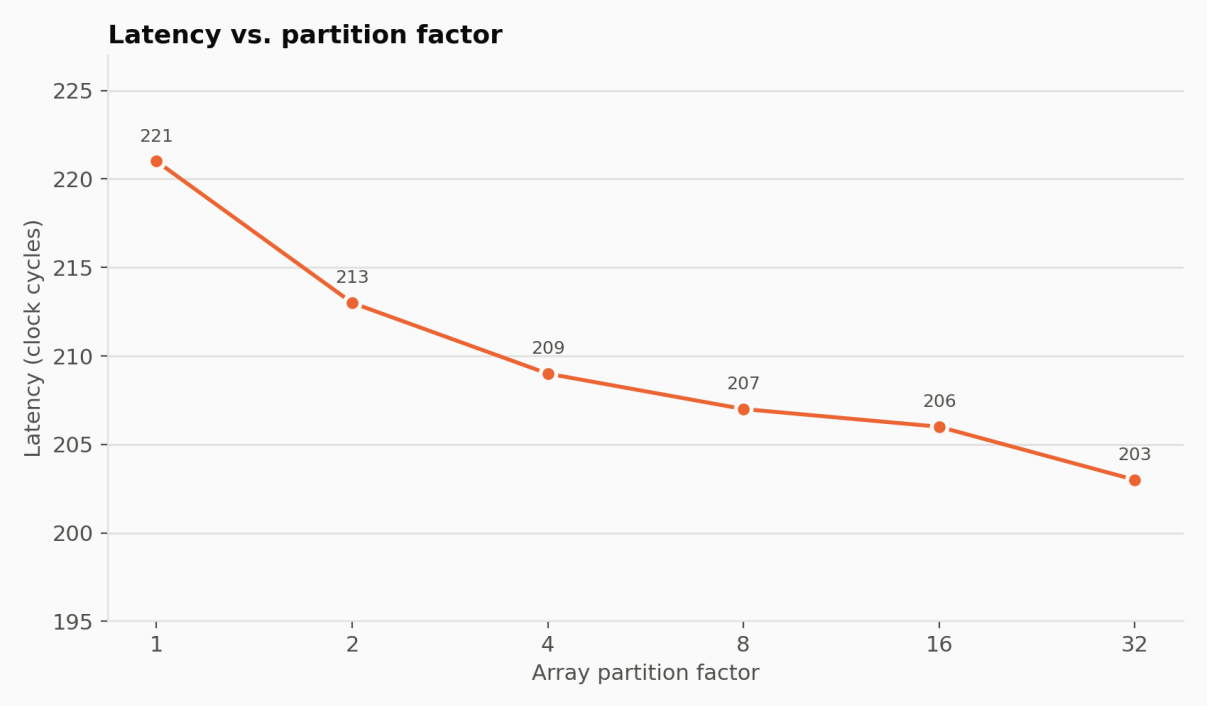
	Factor = 1	Factor = 2	Factor = 4	Factor = 8	Factor = 16	Factor = 32
Latency (clock cycles)	221	213	209	207	206	203
Initiation Interval (II) (clock cycles)	222	214	210	208	207	32
Throughput (DFT/s)	450,500	467,290	476,190	480,769	483,092	3,125,000
BRAM	64	64	64	64	64	64
DSP	640	640	640	640	640	640
FF	59189	59314	59054	58540	57515	61509
LUT	91521	91675	91669	91705	91252	93250

Part (b)



Part (c)


Throughput vs. partition factor
Throughput (DFT/s)
3,500,000
3,000,000
2,500,000
2,000,000
1,500,000
1,000,000
500,000
0
450,450
467,290
476,190
480,769
483,092
3,125,000
1
2
4
8
16
32
Array partition factor


Latency vs. partition factor
Latency (clock cycles)
225
220
215
210
205
200
195
221
213
209
207
206
203
1
2
4
8
16
32
Array partition factor

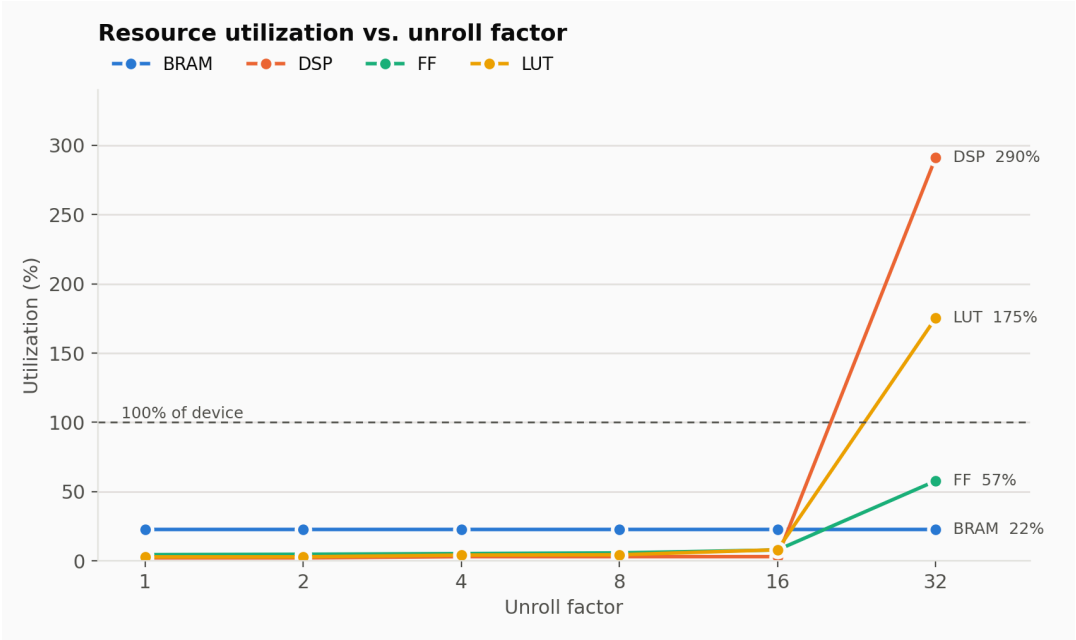
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Question 5 - Loop Unrolling

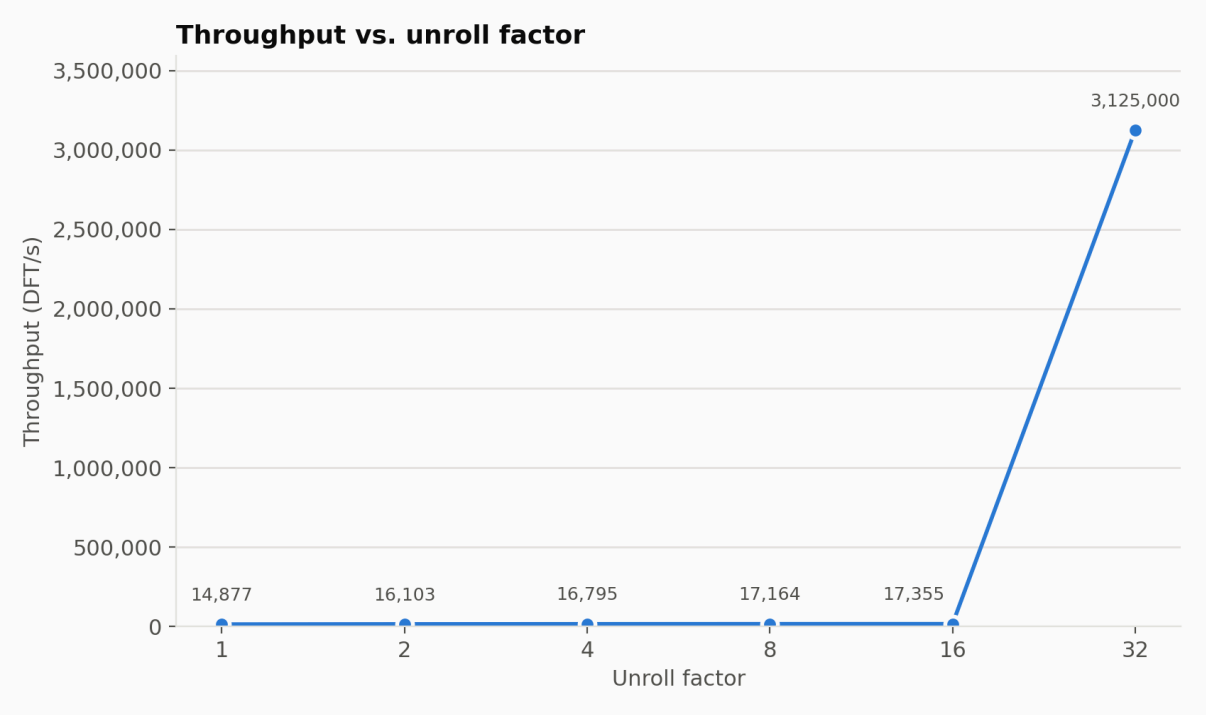
Part (a)

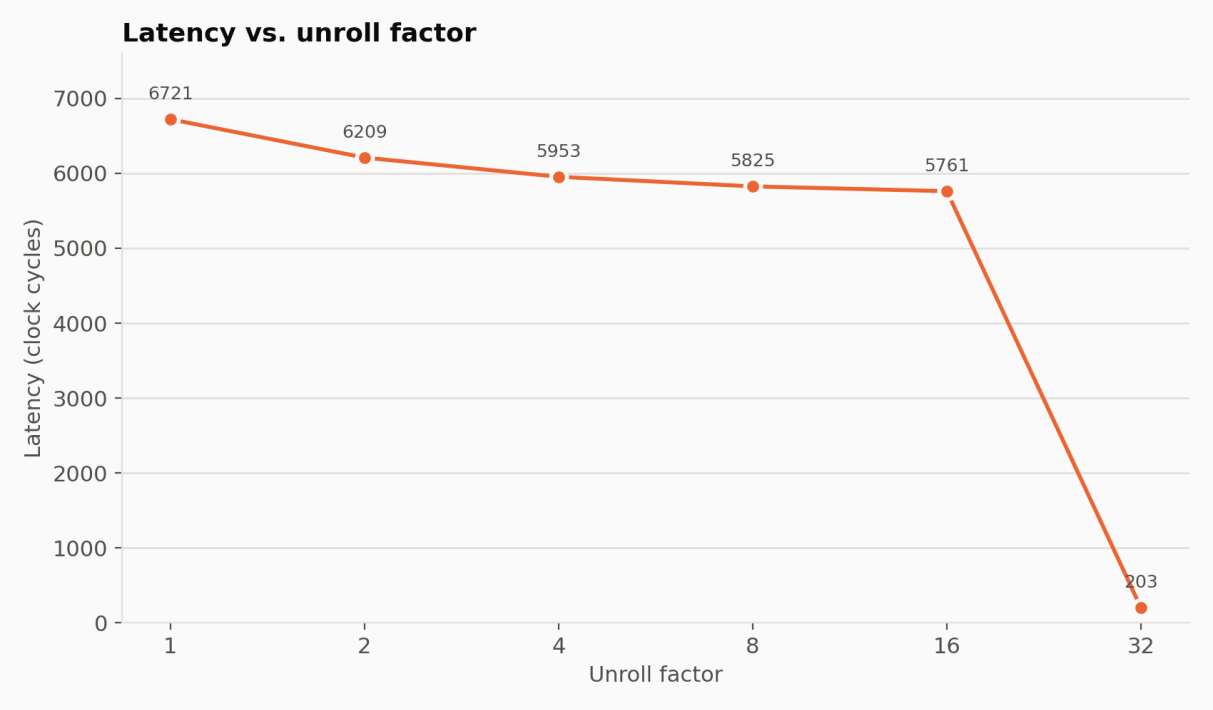
	Factor = 1	Factor = 2	Factor = 4	Factor = 8	Factor = 16	Factor = 32
Latency (clock cycles)	6721	6209	5953	5825	5761	203
Initiation Interval (II) (clock cycles)	6722	6210	5954	5826	5762	32
Throughput (DFT/s)	14,877	16,103	16,795	17,164	17,355	3,125,000
BRAM	64	64	64	64	64	64
DSP	5	5	7	7	7	640
FF	4904	5195	5727	6259	8507	61509
LUT	1596	1621	2295	2413	4339	93250

Part (b)



Part (c)


Throughput vs. unroll factor
Throughput (DFT/s)
3,500,000
3,000,000
2,500,000
2,000,000
1,500,000
1,000,000
500,000
0
14,877
16,103
16,795
17,164
17,355
3,125,000
1
2
4
8
16
32
Unroll factor


Latency vs. unroll factor
Latency (clock cycles)
7000
6000
5000
4000
3000
2000
1000
0
6721
6209
5953
5825
5761
203
1
2
4
8
16
32
Unroll factor

Question 6 - DFT1024 Baseline

Part (a) - Using sin/cos math library

	Baseline
Latency (clock cycles)	55576577 - 56625153
Initiation Interval (II) (clock cycles)	55576578 - 56625154
Throughput (DFT/s)	1.77 - 1.8
BRAM	0
DSP	42
FF	4795
LUT	53200

Part (b) - Using pre-computed sin/cos values

	Table Lookup
Latency (clock cycles)	6291472
Initiation Interval (II) (clock cycles)	6291476
Throughput (DFT/s)	15.9
BRAM	4
DSP	5
FF	1231
LUT	1422

Question 7 - DFT1024 Loop Optimization

	Loop optimization (removed data dependency)
Latency (clock cycles)	1049624
Initiation Interval (II) (clock cycles)	1049625
Throughput (DFT/s)	95.3
BRAM	4
DSP	20
FF	2183
LUT	3380

Question 8 - DFT1024 Best Design

Part (a)

We started with the Q7 implementation. With the data dependency removed, using the pipeline pragma on the inner loop did not give any gains. It turned out that HLS already auto-pipelined it within Q7.
Next, we tried unrolling - but only unrolling did not help. Performance was bottlenecked at writing to the output arrays. After adding cyclic partitioning with the same factor as unrolling, we arrived at our best implementation.

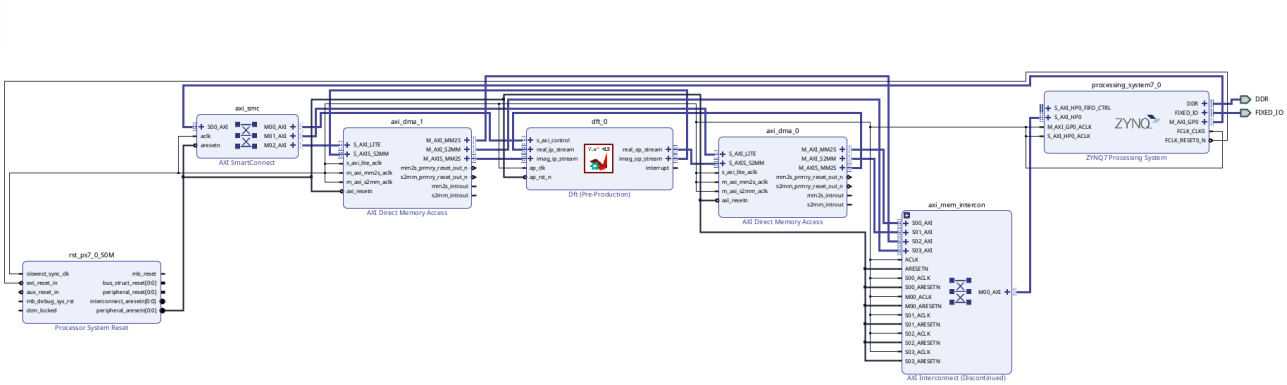
For determining the optimal unrolling and partitioning factor, I did a sweep through 2, 4, 8 and 16. At 16, DSP usage was > 100%.

Part (b)

Latency (clock cycles)	132121
Initiation Interval (II) (clock cycles)	132122
Throughput (DFT/s)	757
BRAM	16
DSP	160
FF	15786
LUT	24750

Question 9 - Streaming Interface Synthesis

Latency (clock cycles)	132151
Initiation Interval (II) (clock cycles)	132152
Throughput (DFT/s)	757
BRAM	52
DSP	160
FF	16003
LUT	25452



GitHub Repo

https://github.com/UCSD-237C-2026/wes-237c-project-3-dft-group-9

Video Demo

https://drive.google.com/file/d/1Pb8xWfjfQvtY4FMwItPTjSRzjrn6njMc